

What's New in Programmable Power Supplies



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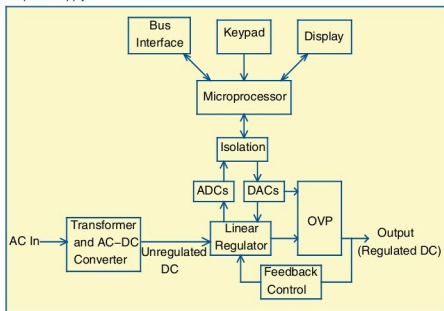
A programmable power supply is a remote-controlled-type power supply that has an analogue input or a digital interface. It is used in a wide variety of applications, including automated equipment testing, crystal growth monitoring, semiconductor fabrication and X-ray generators. It typically employs an integral microcomputer to control and monitor power supply operations. A power supply equipped with a computer interface may use proprietary communication protocols or standard protocols and device control languages such as Standard Commands for Programmable Instruments (SCPI).

Programmable DC power supplies

Power supply manufacturers have to get their devices certified by nationally-recognised primary standards laboratory such as National Institute of Standards and Technology, or simply NIST, or by an internationally-recognised agency such as CSA, UL or VDE.

There are two basic types of programmable DC power supplies that are commonly used: linear and switch-mode.

Fig. 1: Simplified block diagram of a programmable linear DC power supply



Linear power supplies operate by rectifying AC line power to create DC and then filtering and regulating it to produce user-selectable voltage or current levels. These are heavier because the 50Hz or 60Hz transformer and associated filters are physically larger.

Switch-mode power supplies start out the same way, rectifying and filtering AC line input voltage; however these chop the DC into AC. These are significantly smaller, lighter and more efficient than linear ones, and have replaced linear supplies in many applications.

Linear power supplies continue to be a popular choice for test equipment. These are generally durable, accurate and deliver power with little noise. Their simple, direct feedback mechanisms deliver excellent load regulation and overall stability. Fig. 1 shows a simplified block diagram of a programmable linear power supply.

Microprocessors receive input from the user interface or the remote interface. A digital-to-analogue converter (DAC) takes the digital setting and translates it into an analogue value, which is used as reference for the analogue regulator. Setting resolution and accuracy are determined by the quality of this conversion and regulation process. Some common parameters to select a programmable DC power supply are given below:

Resolution and accuracy. Programmable voltage and current settings have resolution and accuracy specifications associated with these. Resolution determines the minimum increment in which the output may be adjusted, and accuracy describes the extent to which the value of the output matches international standards. A DAC with more bits will have more resolution of its output and be able to deliver more distinct values for the control loop to use as reference. Accuracy in